



Supporting Social Play Skill Acquisition and Generalization of Children with Autism Through Video Modeling

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Abstract

This study assessed the effects of portable video modeling on social interactions of four children with autism, three boys and one girl, 4- to 5-year-olds, in preschools in Iceland. Participants were shown 1-min videos on a small handheld device where a peer model initiated social interactions and played with two peers. A multiple-probe-across-participants design showed that video modeling reduced latency to social initiation and increased reciprocal play and language use per 5-min play sessions. Improvements generalized to other peers, a larger group, and to the main classroom and were maintained post intervention. These preliminary findings suggest that video modeling with a handheld device could support the acquisition and generalization of social play of preschool children with autism in natural settings.

Keywords Video modeling · Autism · Generalization · Play · Social skills

Social skills enable people to express their opinions and wishes, connect with other people, and be active participants in society (McGee et al., 1997). Children with autism spectrum disorders (hereafter referred to as autism) often have deficits in social communication and interaction, such as social-emotional reciprocity (American Psychiatric Association (APA), 2013). One effective approach to increasing social skills of children with autism is video modeling (Bellini & Akullian, 2007; Delano, 2007; Fragale, 2014; Shukla-Mehta et al., 2010), but it remains to be explored further to what extent those skills can be applied with novel peers in typical school settings. The aim of this study was to assess the effects of video modeling using a small handheld device on the latency to social initiation, duration of reciprocal play, and language use of children with autism, as well as stimulus generalization across peers and natural preschool settings.

Defining characteristics of children with autism include deficits in social-emotional reciprocity, such as failure to

initiate or respond to social interactions and lack of appropriate back-and-forth conversation skills. They often find it difficult to adjust their behavior to various social contexts, such as participating in reciprocal play, and struggle with making friends (APA, 2013). Social skills typically develop through interacting and playing with peers, giving children opportunities to learn and practice important skills and develop friendships. Children with autism generally have difficulties with imitation, pretend, and reciprocal play (Weiss & Harris, 2001) and show limited interaction with peers during free play (Kroeger et al., 2007). Therefore, effective support is needed for children with autism to acquire play and social skills.

Video modeling is an evidence-based method of teaching skills to children with autism (Fragale, 2014; National Autism Center, 2015; Shukla-Mehta et al., 2010). Video modeling works by showing a short video example of the target behavior, which allows the viewer to imitate the behavior (Nikopoulos & Keenan, 2003). The model in the video (e.g., peers, teachers, parents, or siblings) shows certain target behaviors or language, sometimes with reference to a script. The viewer then imitates the behavior in the video, followed by positive reinforcement which increases the likelihood of the behavior in similar situations (D'Ateno et al., 2003; Reagon et al., 2006). Video modeling can be equally or more effective than live modeling in teaching

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children with autism (Charlop-Christy et al., 2000; Gena et al., 2005).

Numerous studies have shown video modeling to increase a broad range of skills of people with autism (Delano, 2007) such as daily activities (Hong et al., 2015), social initiation (Buggey, 2005; Kabashi & Kaczmarek, 2017; Maione & Mirenda, 2006; Nikopoulos & Keenan, 2003, 2004a, b, 2007; Vidar et al., 2012), play behaviors (e.g. Buggey, 2005; Charlop-Christy et al., 2000; Fragale, 2014; MacDonald et al., 2009; Reagon et al., 2006), and conversational or social skills (e.g. Charlop & Milstein, 1989; Shukla-Mehta et al., 2010). Video modeling has been shown to be effective for children as young as two-years-old (Fragale, 2014). This study focuses on the effects of video modeling on social initiation and reciprocal play in preschool-aged children with autism. Videos with models other than the child viewing the video have been shown to be more effective in promoting social interaction than video-self-modeling with the child him- or herself showing the target behavior (Kabashi & Kaczmarek, 2017). Thus, the focus here will be on video modeling with others as a model.

A few studies have assessed the effects of video-modeling on social skills of preschool-aged children with autism. Maione and Mirenda (2006) investigated the impact of videos with adult models on verbalizations and social interactions of a five-year-old boy with autism in his home. The boy watched three approximately 1-min-long videos on a TV set, each displaying adults playing with one set of toys. Then, the boy was asked to play with these toys, one after the other, with a playmate of similar age for a total of 15 min. Through the video modeling, the boy's social language increased in two of the three activities. His initiations increased from 9–11 per 5 min sessions during baseline to 22–35 after video modeling and his responses increased from 1–4 during baseline to 7–11 after video modeling. For the third activity, both video feedback (referring to the recording of the play) and prompting were needed to produce an increase in social language. Generalization of skills to other settings, people or activities was not assessed, but follow-up probes 7–18 days post intervention indicated that the gains in social language were maintained during the same activities with the same playmates.

Reagon et al. (2006) taught a four-year-old boy with autism pretend play by showing him videos of typically developing peers playing. Each video, that was viewed on a TV screen, showed 20–70 s of pretend play with firefighter, cowboy, teacher, or doctor materials. The video modeling took place in a special preschool for children with autism and greatly added to the boy's skills in pretend play, both regarding language and use of all play materials. Follow-up observations indicated generalization to the home environment and to other family members.

MacDonald et al. (2009) evaluated the effects of video modeling on the reciprocal play skills of 5- to 7-year-olds with autism as they played with typically developing peers. Three videos were produced showing two adults playing with various sets of toys. Video modeling took place in a testing room at the preschool, where the children watched each video twice on a TV set and were then told to play without further instruction or reinforcement. Following video modeling, cooperative play and verbal interactions increased in both pairs of children. In the case of the preschool-aged boy with autism, cooperative play increased from 0.5–6% during baseline to 67–78% after video modeling, and reciprocal verbal interaction chains increased from 0 to 3–7 chains. Improvements in play and verbalizations were still evident at follow-up probes one month after video modeling ended. These findings were replicated by Kouras-sanis et al. (2015) with two 5- to 6-year-olds with autism using 40-s videos showing same-aged (unfamiliar) children playing social games. Through video modeling in a separate room with a TV set and DVD player, along with prompting and praise, game participation increased but verbalizations or generalization to other settings were not assessed.

In most video modeling research, large equipment, such as camcorders, DVD players, and TVs have been applied, often requiring training to take place in separate rooms and/or with a delay between video modeling and opportunities to practice target behaviors in typical settings. The use of handheld devices in teaching children with autism has increased along with technological advances (Kagohara et al., 2013). Portable video modeling has been shown to be equally as effective as that using TVs, but has the advantages of enabling the videos to be easily shown in the appropriate settings, being more motivating for children with autism and more socially acceptable (Miltenberger & Charlop, 2015).

In one of the first studies involving video modeling using handheld devices, Cihak et al. (2010) used video iPods® to teach six- to eight-year-old elementary school students with autism to transition between school activities and locations without problem behaviors. Videos of the students transitioning were filmed with a video camera, edited and then transferred to video iPods®. After being instructed how to operate the devices, students watched each video before transitions and quickly became independent transitioning between activities and locations. Nine weeks after video modeling ended, the students were still able to transition independently without problem behavior. Macpherson et al. (2015) used a small tablet (iPad®) for video modeling of compliment behaviors during outside play. Before intervention, the nine- to eleven-year-old elementary and middle school students with autism rarely used these in play. After video modeling with 20-s video of adults modeling compliments and compliment gestures, the children began to

use them widely in their play, and some generalization to untrained activity occurred.

When children are taught new skills, the goal is for the skills to be used in a variety of settings. Thus, it is important that interventions facilitate generalization of skills across settings as well as over time. Ways to increase the chance of generalization include teaching skills that are naturally reinforced, practicing until high performance levels are reached, and programming common stimuli between the teaching setting and the regular environment (Stokes & Baer, 1977; Stokes & Osnes, 1989). As a result, in video-modeling, generalization can be enhanced by using videos that show participant's peers playing age-appropriately with popular toys and interacting with each other in typical preschool settings. By having children with autism watch their peers play freely with available toys that spur frequent interactions, even a short video can give them numerous examples of social behaviors that are likely to be reinforced in their natural environment. Also, it is advisable to have children practice skills in those settings until they are fully acquired. Subsequently, children need opportunities to apply the skills in gradually more complex settings. Accordingly, participants in this study, watched videos of models showing behaviors that are likely to be reinforced by peers in typical preschool settings, and after acquiring those behaviors, practiced applying them to novel peers and larger groups.

Most previous research on video modeling with preschool-aged children has taken place in settings other than typical diverse preschool classrooms, such as home settings (e.g., Maione & Mirenda, 2006), special schools (e.g., Reagon et al., 2006), or separate rooms (e.g., MacDonald et al., 2009). This study was conducted in typical preschool settings, using a small handheld device for both recording and playback of videos. The aim was to explore the effects of video modeling on the latency to social initiation, duration of reciprocal play, and language use of preschoolers with autism during play with peers in gradually more complex preschool settings. More specifically, generalization to novel peers, a larger group of peers, and to a large group of children in the main classroom was assessed, as well as maintenance of intervention effects.

Method

Participants

Four children with autism, from four public preschools in the capital region of Iceland participated in the study. They were nominated for participation by educational consultants, based on intake criteria of: (a) being diagnosed with autism and significant delay in verbal and social skills, (b) being 4 to 5 years old, (c) lacking social initiation, and (d)

needing extensive, continuous adult prompting to engage in interaction or reciprocal play with peers.

Laura was a 5.4-year-old smiley, cheerful, and cooperative girl, with a diagnosis of atypical autism and intellectual disability from the *State Diagnostic and Counseling Centre* (SDCC). Her intellectual development was measured at two standard deviations below the norm, as reflected in a score of 70 on the *Bayley Scales of Infant and Toddler Development* (Bayley, 2005). Her motor skills and language development were also below average. Laura received support for eight hours a day and had received early intensive behavioral intervention (EIBI) since the age of 3.1. According to her teacher, she was able to imitate and watched other children play, but rarely joined them and showed limited interaction and endurance in reciprocal play.

John was a 5.5-year-old lively and energetic boy, diagnosed with autism, intellectual disability, and attention deficit hyperactivity disorder (ADHD) from SDCC. His intellectual development was estimated at two to three standard deviations below the norm, as reflected in scores of 53 to 66 on the *WPPSI-III* (Wechsler, 2002). John's behavior scored 31, in the mild to moderate range on the *Childhood Autism Rating Scales* (CARS) (Schopler et al., 1988). John received support eight hours a day with TEACCH (*Treatment and Education of Autistic and related Communication-handicapped Children*). He lacked endurance and concentration during most activities and found social interaction difficult. John joined the study a little later than the others, reflected in a later start of baseline measurements.

Solvi was a 5.4-year-old sweet, quiet boy, with a diagnosis of autism and intellectual disability from SDCC. Specifically, his language development was significantly impaired. With a score of 41 points on CARS, he had severe symptoms of autism. Solvi received eight hours of support a day and had received EIBI since the age of 3.3. He was able to label most items and had basic imitation skills. He watched other children play but rarely interacted with them, often showed defiance when prompted to do so and challenging behavior in social or group settings.

Thor was a 4.10-year-old calm boy with great interest in cars and animals. He had diagnoses of autism and language impairment from SDCC. Thor's cognitive development was measured at one standard deviation below average as reflected in a score of 81 on the Bayles scales. His language comprehension and expression fell two standard deviations below average, as reflected in scores of 63–69 on a standardized language development test. His autism symptoms measured 29 on CARS. Thor received seven hours of support a day and had received EIBI since he was 3.6. Thor's main challenges involved social interaction and playing with peers.

Settings

All baseline and intervention sessions took place in each participant's general preschool classrooms. Laura's preschool had almost 50 children in two classrooms, thereof 25 in Laura's classroom. Observations took place in one of the three rooms of the classroom. The room most often used was 7 m² and contained one table with six chairs, shelves with various materials in containers, and a mirror on one wall. Generalization assessments to the largest group of children ($n = 12$) took place in the largest room, that was 26 m² with two, eight-person tables.

John's preschool included 110 children in five classrooms, thereof 25 in John's classroom. Observations took place in one of two rooms. It was 20 m² and contained a table with six chairs and a play kitchen with a sink, a cabinet, and a stove. There were two rugs on the floor and along the walls were shelves with various play materials and napping mattresses.

Solvi's preschool had 94 children in five classrooms, thereof 25 in Solvi's classroom. Observations most often occurred in a 6 m² room with a sofa and a high mounted shelf with a music player and books. Generalization assessments to the largest group of children ($n = 14$) took place in the main room. That room was 20 m², with two, six-person tables, a computer table, two floor mats, and shelves along the wall with various items and materials.

Thor's preschool had 86 students in four classrooms, thereof 22 in Thor's classroom. All observations took place in one of the two rooms in the classroom. It was 23 m², with two, six-person tables, a sofa bed, and shelves with various toys and books.

Play materials

Prior to onset of baseline measures, participants' peers were observed playing with different materials in each preschool to choose those that spurred the most interactions between children of the participants' developmental stage. Another criterion was that each child could have two to four items (animals, vehicles, people, etc.) for play. During generalization assessment the number of play items was increased according to group size. In two cases, new toys were bought. Each play set was reserved for use throughout the study. Laura and her peers played with people and animals in a three-story, wooden house with open sides, a swing, a rope ladder, and a viewing platform that turned. John and his peers played with a Noah's Ark that had a wooden bow, a gangplank, water buckets, barrels, lanterns, shelves, people, animals, and food for the animals. Solvi and his peers used a Playmobil® farm playset that was hinged in the middle and had a tractor, people, and animal stalls. Thor and his peers played with a Lego Duplo® farm set with a stable, animals,

a barn with a hayloft, a crane to lift hay, a tractor, a car, a milk tank, and a fence.

Equipment

A small handheld device, iPod touch® with video recording and playback capabilities was used to record and show participants videos for video modeling on a 3.5 inch screen. Recordings for measurements were made with a Canon Legria® HF 41 camcorder on a tripod and a microphone connected to the camcorder with a 5-m extension cable. The microphone was taped to the floor 70–100 cm from the playsets to maximize sound quality. It was hidden behind curtains or furniture to minimize disturbance. A stopwatch was used to measure latency to social initiation and play duration. A counter was used to measure the number of words, with each button push indicating a word.

Experimental Design

A single-subject multiple-probe across participants design (Horner & Baer, 1978) was used to evaluate the effects of video modeling on the social skills of participants and generalization to novel and larger groups of peers. A multiple-probe design is a variation of a multiple-baseline design, where intervention is implemented sequentially, after varying lengths of baselines, with the goal of demonstrating changes in the dependent variables only when the independent variable is manipulated. A functional relationship and the magnitude of effect is determined through visual inspection of the immediacy, level and pattern of change in the dependent measures compared to baseline (Ledford & Gast, 2018). Intermittent probes were conducted in this study to minimize the times that children with autism were instructed to play with other children without intervention. Intermittent probes are considered appropriate when extended continuous baseline measures are impractical, may prove reactive and/or when stability in dependent variables can be assumed (Horner & Baer, 1978). Dependent variables were assessed during baseline (A), intervention (B), generalization phases, and at follow-up 6 to 7 weeks after end of intervention.

Independent Variable

The independent variable was video modeling using an iPod touch® handheld device. Each participant was shown a 1-min long recording of three peers from his/her classroom in reciprocal play. Peer models were identified by teachers as socially competent, friendly and helpful. They were the same age as participants, but their previous interactions were little if any due to participants' lack of social and play skills. During visits of the second author to the preschools in the week prior to recordings, peer models were allowed to play

with the toy sets (farm, treehouse, or Noah's ark) selected and reserved for use in the study, and encouraged to create stories, actively interact and describe their actions while playing. They also practiced the sequence that was to be recorded: two of them playing together on the floor when the third (the most talkative) enters and asks "Can I play?" Then they all played actively together for the rest of the minute. The second author created all four videos, one for each participant, recorded in the same rooms and with the same two classmates playing as present during baseline and intervention phase.

Response Definitions and Measurement

Partly based on research of Nikopoulous and Keenan (2004) three dependent variables were defined and assessed from 5-min recordings of play: latency to social initiation, duration of reciprocal play, and frequency of words during play. *Latency to social initiation* was defined as the time in seconds from when a participant entered the room until he/she began interacting with a peer verbally (with words or play sounds) or with motions such as touching or sitting down next to a peer, or by handing or showing a toy. If no initiation occurred, a 300 s latency was recorded. *Reciprocal play* was measured as the total time that the participant engaged in positive social interaction with his or her peers, such as verbal expressions (with words or play sounds) with or without a toy, gestures such as touching and looking at, sitting next to, showing or handing a toy to a peer. If a participant took a toy from a peer, pushed, or interacted in another negative way or 10 s passed without positive social interaction, the recording of reciprocal play was paused. *Language use in play* was measured by the number of words a participant directed towards peers, toys, and/or were related to the play during a 5-min observation period.

Interobserver Agreement

Two observers (the second author and an independent researcher) practiced measuring the dependent variables until they reached 90% agreement. Interobserver agreement

(IOA) was evaluated in 33% of the video-taped observations for each dependent variable in all experimental phases (see Table 1). Observers watched the tapes non-simultaneously, and independently scored the dependent variables on a custom-made recording sheet. If there was less than a 2-s difference between their duration or latency recordings, they were considered to be in agreement. Frequency and overall IOA were calculated by compiling measurements where observers agreed, dividing those by the total of measurements and multiplying by 100%. On average, IOA was 96.5%.

Procedure

This study was conducted in accordance with act 77/2000 on the Protection of Privacy as Regards the Processing of Personal Data and registered at the Icelandic Data Protection Authority. All relevant approvals and consents were obtained from the municipality, preschool directors and staff, and parents of participants, peers and video models. The second author conducted the study along her work as educational consultant, visiting each preschool three times prior to onset of baseline to train support persons in their role, choose appropriate sets of toys, select and become acquainted with peer models, train them in their role, and record videos for each participant. Peers were instructed to play and not to initiate contact with the participant when he/she entered the room, but to respond positively to all attempts of him/her to interact or play. They were instructed to be "good role models" and support participants' playing, for example by speaking slowly, explaining, and encouraging play and friendship skills. Preschool staff was trained to show participants the video, say "Go play", and not to give any other prompts or feedback before, during or after sessions. No other adult interventions took place during observation sessions. Staff was also asked to divert possible questions from participants and peers regarding the study.

Observations took place at a fixed time of day for each participant, according to the multiple-probe design. After baseline, sessions were scheduled daily, but some gaps occurred due to field trips, illness of participants, peer models or staff,

Table 1 Interobserver agreement of dependent variables by research phase

	Latency of social initiation	Duration of reciprocal play	Language use
Baseline	88%	81%	94%
Intervention	100%	94%	88%
Generalization to novel peers	100%	100%	100%
Generalization to larger group	100%	100%	100%
Generalization to main classroom	100%	100%	100%
Follow-up	100%	100%	100%
Average	98%	96%	97%

and so forth. All observation sessions were recorded for 5-min, starting when the participant entered the room.

Baseline

Prior to each observation session during baseline (and generalization) phases, each participant's support person was asked to take the participant to the entrance of the preassigned room and say "Go play" without further prompting.

Intervention

Each participant's support person was instructed to take the participant to the entrance of the preassigned room, show him/her the video once in the handheld device, and say "Go play" without further prompting. The same video was used throughout the phase.

Stimulus Generalization

As during baseline, the support person was asked to take the participant to the entrance of the preassigned room, and say only "Go play" but now with two novel peers, then with five to six peers, and finally with 8–14 peers in the largest room in their classroom. All conditions, except the one with novel peers, included the two peers from the video that were present in all baseline and intervention sessions. Other peer volunteers were also asked to not initiate contact with the participant but to respond positively to all attempts of him/her to interact or play. Six to seven weeks after the end of intervention and four weeks after generalization measurements, follow-up observations assessed maintenance of increased skills.

Social Validity

At the end of the study, preschool staff was interviewed about their perception of the intervention.

Data Processing and Analysis

Data from the observations were entered into Excel sheets and set up in figures and tables. Functional relationships and effects of video-modeling were determined through visual analysis of the immediacy, level and pattern of change in the dependent measures compared to baseline (Ledford & Gast, 2018). Interviews were recorded, transcribed, and thematically analyzed.

Results

The results for each research question are presented separately.

Latency to Social Initiation

Figure 1 shows the latency to social initiation for each participant during baseline, intervention, generalization, and follow-up phases. All participants showed positive changes as soon as video modeling began. The average latency decreased from 138 to 27 s after intervention.

During baseline observations, Laura took 16–83 s to initiate interaction, with an average of 51 s. After video modeling, latency decreased to 2 s, which held throughout the intervention phase. During generalization phases with new or more peers present, Laura took 4 s on average to initiate interaction, just like during follow-up sessions six weeks after the end of intervention.

John did not initiate interaction in five of the baseline observations, but in four others it took him 22–150 s, and the overall average was 204 s. During the intervention phase, the latency showed a declining trend and reduced to 30 s on average. When other or more children were added, John took 3–11 s for initiation, on average 5 s, the same as during follow-up.

Solvi only initiated interaction once during baseline observations, after 19 s, resulting in an average latency of 274 s. After intervention, the level of latency dropped and showed a decreasing trend with some variability and an average of 73 s. Towards the end of the intervention phase, a stable low level of latency below 20 s was achieved. When generalization to novel peers was assessed, Solvi took much longer to initiate, 139 s on average, but with a declining trend. With a larger group, the latency was 10–37 s, but no interaction occurred in the largest group in the main classroom. At follow-up, Solvi initiated interaction after 26 s.

During baseline, Thor initiated interaction after 8–34 s, 22 s on average. After intervention, Thor's latency decreased to 1–6 s, 3 s on average. During generalization phases, Thor's initiation latency was 2–22 s, or 9 s on average and at follow-up, 3 s.

Reciprocal Play

Figure 2 shows the duration of reciprocal play with peers for each participant during baseline, intervention, generalization, and follow-up phases. Laura's play with peers during baseline was variable, lasting 77–230 s out of 300 s, or 135 s on average. After intervention, play duration showed a sharply increasing trend and soon became stable at 300 s, averaging 278 s overall. During generalization and follow-up phases, Laura's reciprocal play averaged 296 s.

John showed no reciprocal play in five of the nine baseline observations. When he did, he played 15–102 s, with an average of 22 s. During intervention, John's reciprocal play showed a positive trend, increasing from 126 to

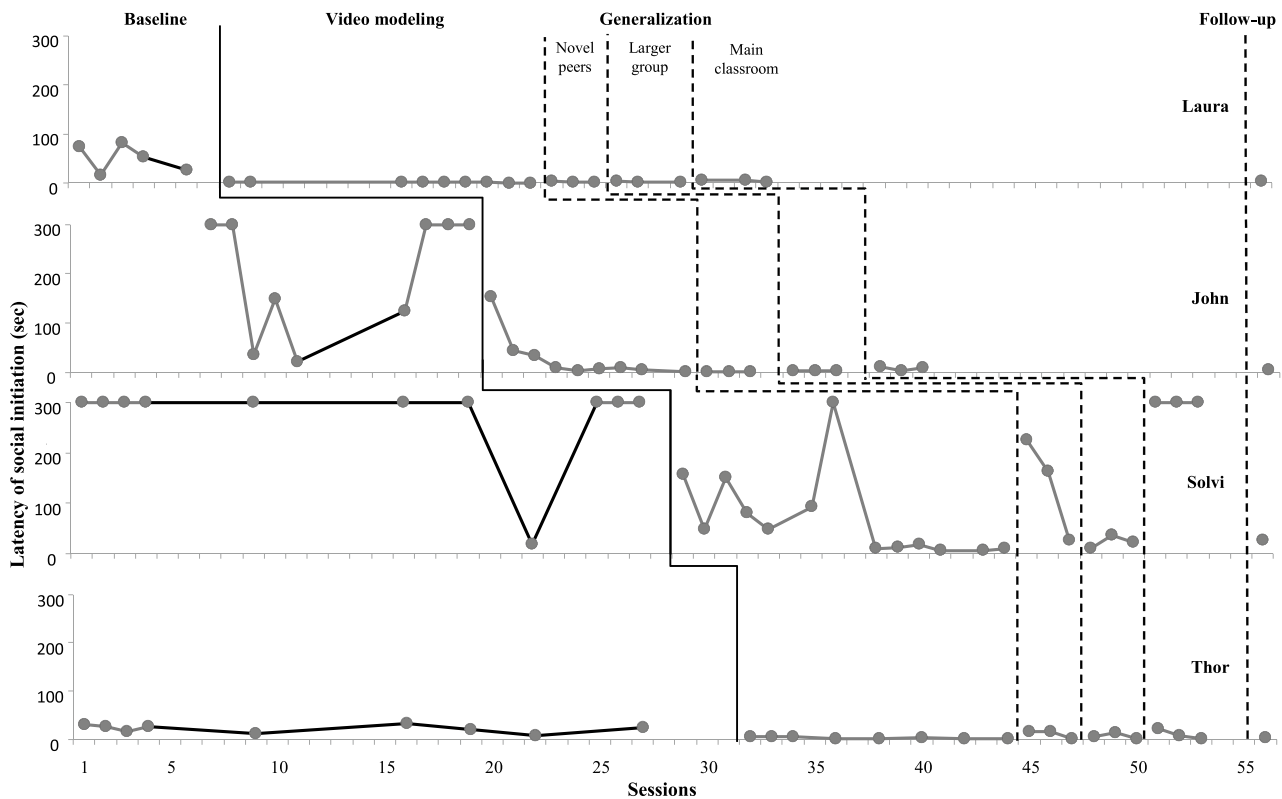


Fig. 1 Latency to social initiation by participant during baseline, video-modeling, generalization (to novel peers, a larger group of peers, and to the main classroom), and follow-up phases

298 s, with an average of 264 s. During generalization and follow-up phases, John's reciprocal play averaged 286 s.

Solvi only took part in reciprocal play once during baseline for 85 s, averaging 8 s. After intervention, Solvi's play duration was variable, starting with a higher level but decreasing trend, progressing to an increasing trend, averaging 100 s overall. When generalization to novel peers was assessed, Solvi's play initially dropped but showed an increasing trend. When more peers were added to the group, Solvi's play showed a decreasing trend and in the main classroom with the largest group, Solvi did not engage in any reciprocal play. On average, Solvi played reciprocally for 103 s during the generalization phases and for 111 s during follow up.

Thor's reciprocal play lasted between 74 and 284 s during baseline, averaging 236 s. With video modeling, the duration of play became stable around an average of 297 s. When measuring stimulus generalization, his play duration remained high and stable, averaging 290 s. Follow-up measurements averaged 297 s.

Language Use in Play

Figure 3 shows the number of words each participant used playing with peers in 5-min observations during baseline, intervention, generalization, and follow-up phases. Prior to intervention, Laura's language use was low and stable with 0–9 words (averaging 5), which increased to 15–44 words (averaging 30) after video modeling. During generalization assessment, Laura used 21–46 words (averaging 32) with novel peers, 28–145 words (averaging 76) with the larger group, 29–33 words (averaging 32) with the largest group in the main classroom, and 77 words at follow-up.

John used 0–28 words (averaging 7) in the 5-min observations during baseline. With video modeling John's speech showed a positive trend, with 24–126 words (averaging 69) during 5-min sessions. During generalization assessment, John's language use showed a negative trend with novel peers, with 39–104 words (averaging 79), but a positive trend in the larger group, with 33–99 words (averaging 74). In the largest group, the level of language use decreased to

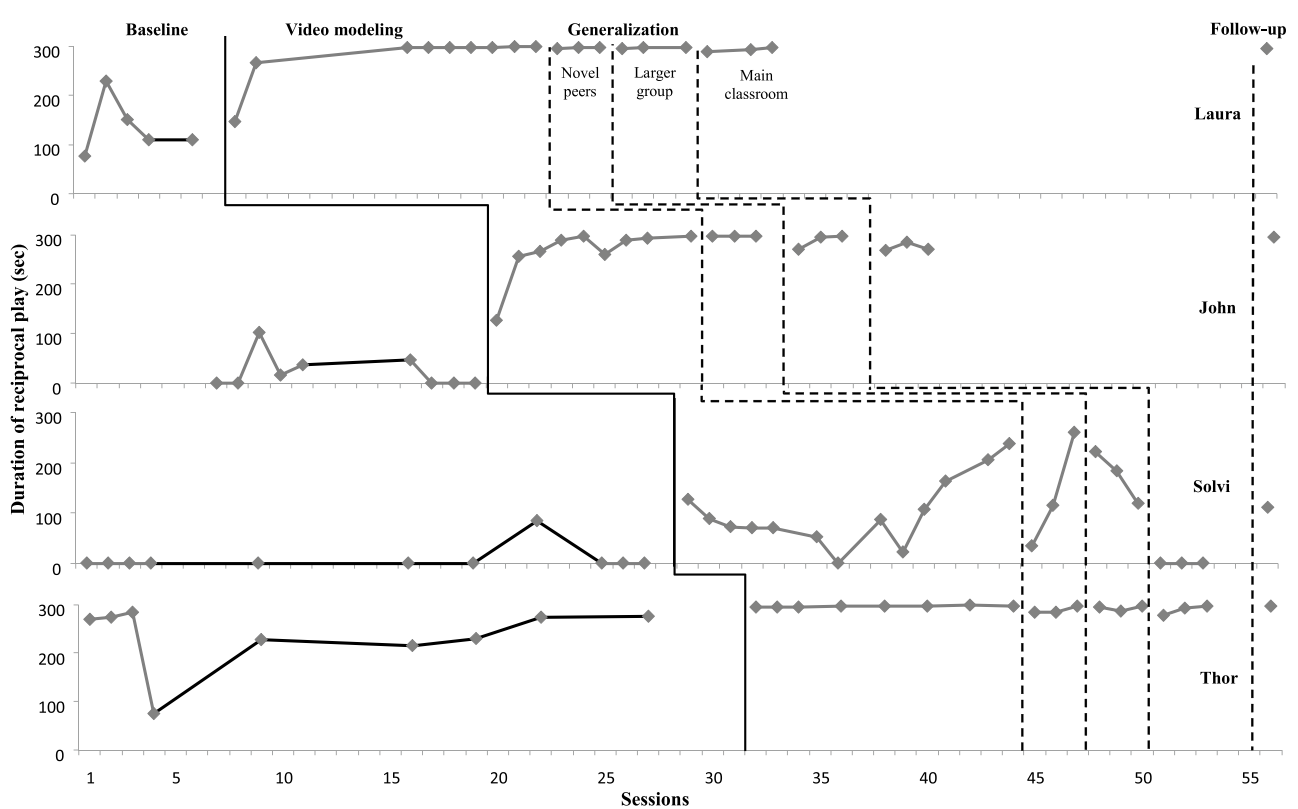


Fig. 2 Duration of reciprocal play by participant during baseline, video-modeling, generalization (to novel peers, a larger group of peers, and to the main classroom), and follow-up phases

16–55 words (averaging 36), but at follow-up, John used 110 words. Solvi used 0–16 words (averaging 1.5) during baseline, but with video modeling his language use showed a positive trend with 0–43 words (averaging 13). During generalization assessment, Solvi used 8–20 words (averaging 12) with novel peers, 11–31 words (averaging 13) with the larger group, but no words in the main classroom. At follow-up Solvi used 7 words. Thor used 5–35 words (averaging 19) in the 5-min baseline observations, but with video modeling his language use showed a positive trend with 27–146 words (averaging 82). During generalization assessment, Thor used 103–153 words (averaging 127) with novel peers, 69–149 words (averaging 119) in the larger group, and 70–222 words (averaging 128) in the main classroom. At follow-up Thor emitted 65 words during the 5-min observation.

Social Validity

When interviewed about their perception of the intervention, preschool staff described positive views. All interviewees considered video modeling to be an effective and simple intervention and expressed that they would like to apply it with other children. They reported that participants now more often asked peers if they could play, showed greater

playfulness and increased positive interactions. According to interviewees, peers who participated in the study also seemed to understand participants with autism better, seeing them more as peers, expecting more interaction from them, and enjoying their company more than before the intervention. All interviewees concluded that they would recommend video modeling to their colleagues.

Discussion

The results showed positive effects of video-modeling on the social interactions of the four preschoolers with autism in this study with their peers. After intervention, participants initiated interaction with shorter latency, played longer with their peers, and used more words while playing than before. The increased skills generalized to new playmates, larger groups, and the main classroom in most instances. These preliminary results indicate that video-modeling with a small handheld device could be an effective way of promoting play and social interactions of preschool children with autism.

Video modeling shortened the latency to initiating interactions at the beginning of play sessions for all participants

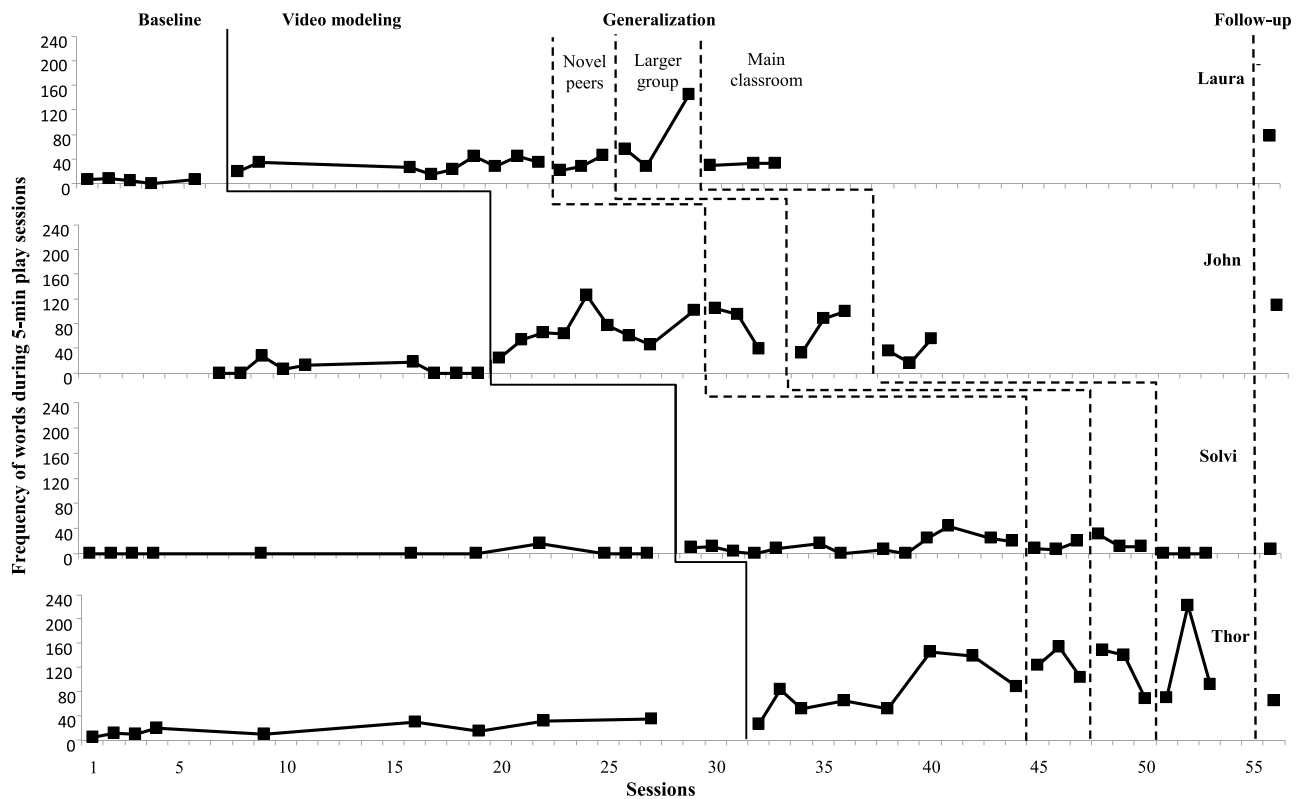


Fig. 3 Language use in play sessions by participant during baseline, video-modeling, generalization (to novel peers, a larger group of peers, and to the main classroom), and follow-up phases

from an average of 138 s to 37 s. The latency decreased most in the case of John. He greatly enjoyed the video and often pointed to the model, asking “Am I this one?” Prior to intervention, Solvi only initiated interaction once and seemed uncomfortable in the situation. After video modeling, he almost always initiated interaction and seemed more at ease and happier at play. The shorter latency to initiation generalized to new peers and a larger group in all cases, and to the largest group in the main classroom in all cases except Solvi.

The decreased latency to social initiation following video modeling is in line with results from previous research (Kabashi & Kaczmarek, 2017; Maione & Mirenda, 2006). Most of this research has taken place in elementary schools, using TV sets for video modeling (Buggey, 2005; Nikopoulos & Keenan, 2003, 2004a, b, 2007; Vidar et al., 2012). The current results indicate that 1-min videos with peer models shown on small handheld devices could have a significant impact on the social initiations for children with autism in a general preschool setting.

The duration of reciprocal play increased for all participants, on average from 100 to 235 s during each 300-s observation. After video-modeling, participants gradually played longer and more actively, and, in the opinion of the observers, seemed to enjoy interacting with their peers. It

should be noted that peer models were particularly selected because of their helpfulness, and received instructions on how to play with the participants, for example by speaking slowly, explaining, and encouraging to take turns. This careful selection and preparation probably enhanced the participants’ chances of having positive interactions, increasing the natural reinforcement of their initiations and play behavior. Attempting to play with a more socially diverse group initially, would most likely have led to less reinforcement and slower acquisition of social skills. Thus, participants were set up for success and practiced social interactions in a safe environment, which probably played a role in the generalization to novel peers and larger groups. It remains to be explored whether similar effects can be achieved without specifically selecting socially apt peers for the initial practice of social interactions during the video modeling phase.

Increased reciprocal play is in accordance with earlier research showing video modeling improving the play skills of children with autism (Fragale, 2014). Most of the research has targeted children in elementary schools (Buggey, 2005; Charlop-Christy et al., 2000; Nikopoulos & Keenan, 2003, 2004a, b, 2007; Vidar et al., 2012). In the studies targeting preschool-aged children, video-modeling has been applied at home (Reagon et al., 2006), in a separate room at a treatment

center (Kourassanis et al. (2015), or in a testing room (MacDonald et al., 2009). In all studies DVD or VCR players and TVs were used for video modeling. The present findings indicate that short videos, recorded on a small handheld device, with peer models, might be helpful in extending the reciprocal play of children with autism in natural preschool settings.

Language use of participants increased on average from 8 to 49 words per 5-min observations after video modeling. Laura's, John's, and Thor's increases happened quickly, but Solvi took longer and mostly talked for the play figures. The increased language use generalized to novel peers, a larger group, and to the main classroom for all participants but Solvi. Anecdotal reports of preschool staff also indicated increased social language use in other settings. For example, John was reported to be more outgoing, frequently interacting with peers and staff.

Participants' increased language use is in agreement with previous research showing positive effects of video modeling on communication skills of children with autism (Charlop & Milstein, 1989; MacDonald et al., 2009; Shukla-Mehta et al., 2010). Most often, videos of adult models have been shown on TVs, as in MacDonald et al.'s study, in which three videos were shown twice per session, much more frequently than in the current study. More extensive modeling may be needed to teach complex skills, like scripted play scenarios and verbalizations, compared to unrestricted expressions and play, that were targeted in this study.

The goal of intervention is for learned skills to maintain and generalize to other circumstances in the daily lives of learners (Charlop-Christy et al., 2000). In this study, generalization was promoted by programming common stimuli (e.g., using peers and playsets from participants' classrooms), introducing natural maintaining contingencies (e.g., modeling behaviors likely to be reinforced and asking peers to be responsive to participants' efforts), training loosely (e.g., using wide definition of appropriate initiation, play and language use), and indiscriminable contingencies (e.g., unpredictable reinforcement on behalf of peers) (Stokes & Baer, 1977). After initiation and play duration had increased with video modeling, participants' increased skills generalized to new playmates and to a larger group. All participants but Solvi also continued to initiate and play with peers in the largest group in the main classroom, despite a noisier and more complex setting. However, their language use decreased somewhat, indicating that smaller peer groups are better suited for promoting language use in children with autism.

Solvi stood out from other participants. His level of autism was more severe and his progress was slower and required considerably more sessions to shorten latency to initiation and increase reciprocal play. He had not reached a high and stable level of reciprocal play when intervention

ended, perhaps leading to less generalization and maintenance than observed with other participants. Perhaps additional prompts, such as verbal or visual reminders to play and talk, and/or more direct reinforcement of such behaviors could have spurred Solvi's progress in social play (Fragale, 2014; Maione & Mirenda, 2006). Also, he had according to staff "no good experience" with bigger groups of children, and there had recently been staff changes in the main classroom. Possibly changing the settings more gradually could have led to better generalization and maintenance. Maybe just adding one new peer at a time, and preparing new peers in a similar manner as the ones playing with him during the video-modeling phase could have made the transitions more successful for Solvi. Further research is needed to explore how video modeling can be individualized to better meet different needs of children with autism and to maximize chances of generalization. These results are yet another reminder that even generally effective interventions are unlikely to be successful for every child with autism and underscores the importance of data-based decision making to tailor interventions to different needs (Ledford et al., 2018).

The stimulus generalization and maintenance observed in this study is in accordance with previous studies of video modeling (Charlop & Milstein, 1989; MacDonald et al., 2009; Maione & Mirenda, 2006; Nikopoulos & Keenan, 2003, 2004b, 2007; Reagon et al., 2006). Most often older children have been targeted, in segregated settings, with older models, and/or more cumbersome equipment. Preschool-aged children, however, have rarely been targeted. Two studies reported maintenance of play-related verbalizations of 5-year-olds over 18 days (Maione & Mirenda, 2006) and one month (MacDonald et al., 2009), but generalization to novel peers or settings was not assessed. In the study of Reagon et al. (2006), the increased play skills of a 4-year-old maintained and generalized from his special school to new playmates at home after video modeling with his older brother as the model. The present study contributes to the literature by assessing generalization to large groups in general preschool classrooms after video modeling with a small handheld device, using preschool peer models. Peer models have several advantages when teaching social skills and can support generalization (Hughes et al., 1995). Peers are part of the natural school environment and therefore "common stimuli" in intervention and many daily settings. They also model typical play and conversational skills that are likely to be reinforced throughout the day. In addition, when peers receive specific training as role models and educators for their peers with autism that can help them further in interactions after intervention ends. This aligns with staff reports in the present study, describing how playmates showed better understanding of their peers

with autism, asked them to play more often, and prompted more interaction during play. This could have supported generalization into new settings and over time. Further research is needed to explore whether peer models in video modeling support generalization more than adult models.

This study has several limitations, including the lack of direct assessment of treatment fidelity and social validity. Anecdotal evidence indicated that teachers implemented the simple intervention correctly, with the exception of one or two instances where they attempted to provide additional prompts but were stopped by the researcher. Future research should include direct assessment of treatment fidelity as well as social validity to shed more light on teachers' application, acceptance, and perceived benefits of video-modeling in natural preschool settings. Also, implementing a multiple-probe design with participants in four public preschools, alongside work responsibilities of the second author as educational consultant, proved challenging, and it is unclear how different phase lengths and gaps in measurement, such as between baseline and intervention phases, affected the findings. In any case, as with all single-subject design studies, the findings have limited generalizability and call for direct and systematic replication. Lastly, more thorough follow-up measures would be useful to assess the long-term effects of video-modeling on social skills of children with autism. Including broader outcome measures in future research could shed light on whether continuing application of video-modeling to address various skills deficits of children with autism can support their overall inclusion in general education settings, friendship development, and general well-being.

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Author Contributions Both authors collaborated on the study conception, design, and data analysis. Material preparation and data collection were performed by TG under the supervision of ALP. The manuscript was written by ALP. Both authors approved the final manuscript.

Data Availability Data from the study were previously published in Icelandic (Gudmundsdottir & Petursdottir, 2017). The data that support the findings of this study are available from the corresponding author upon request.

Declarations

Conflict of interest We have no known conflict of interest to disclose.

Informed Consent Written informed consent was obtained from teachers and parents of children for participation in the study and for publication of the research findings in journals.

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